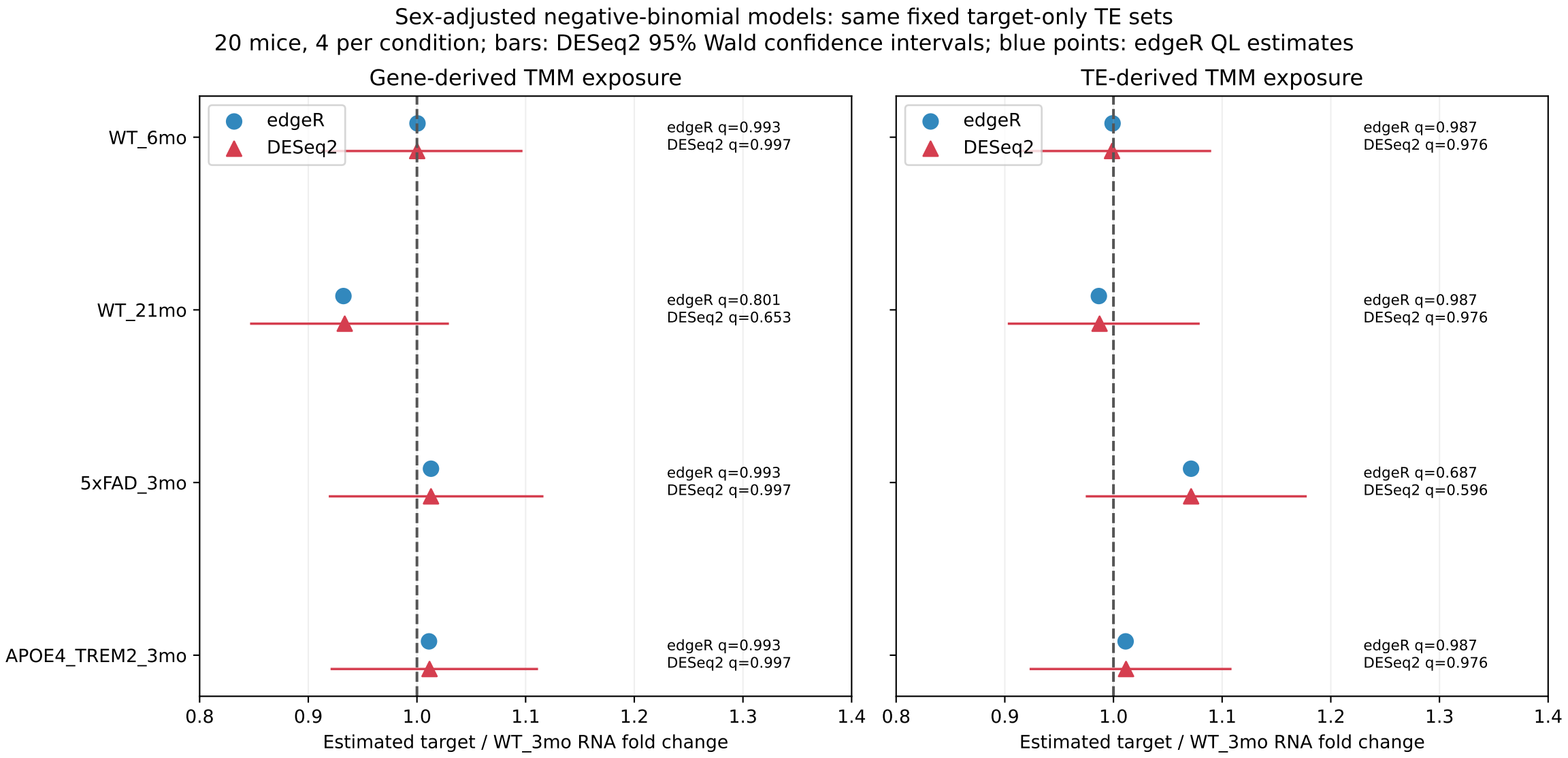
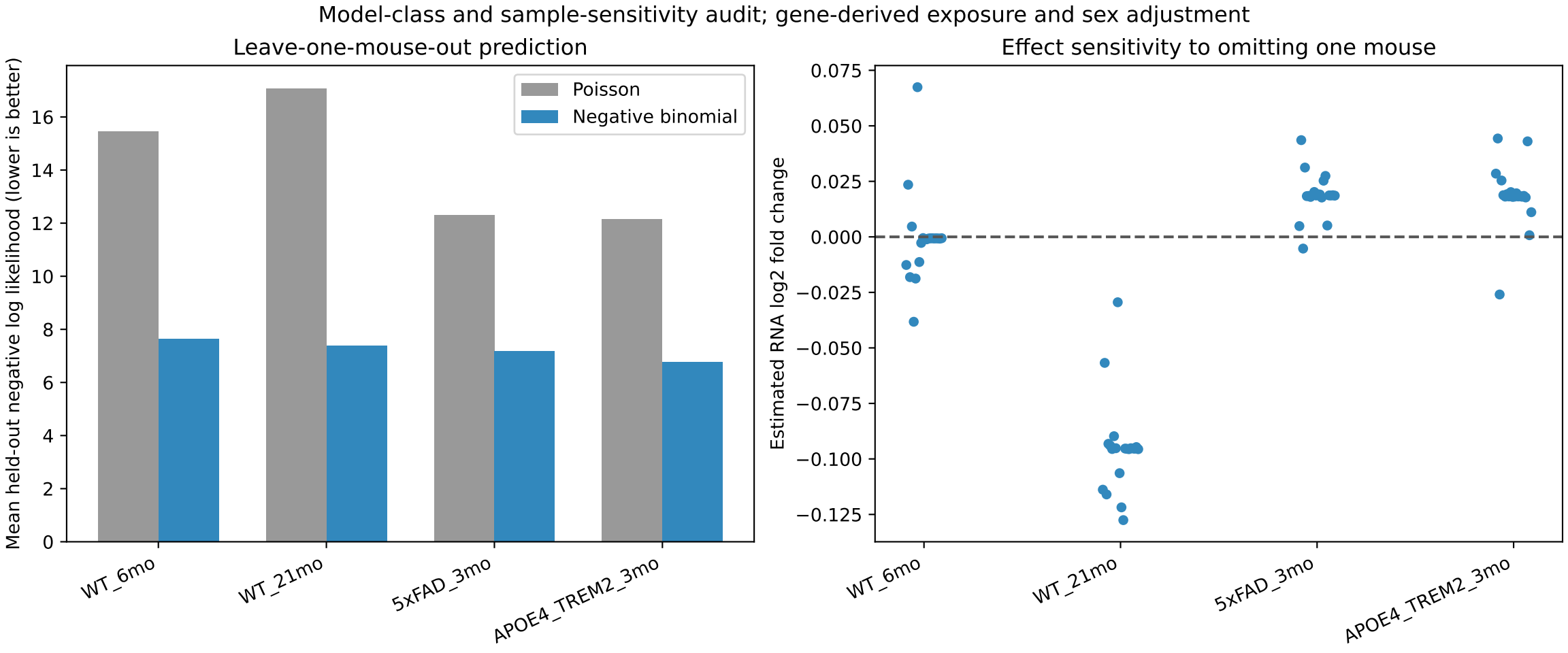
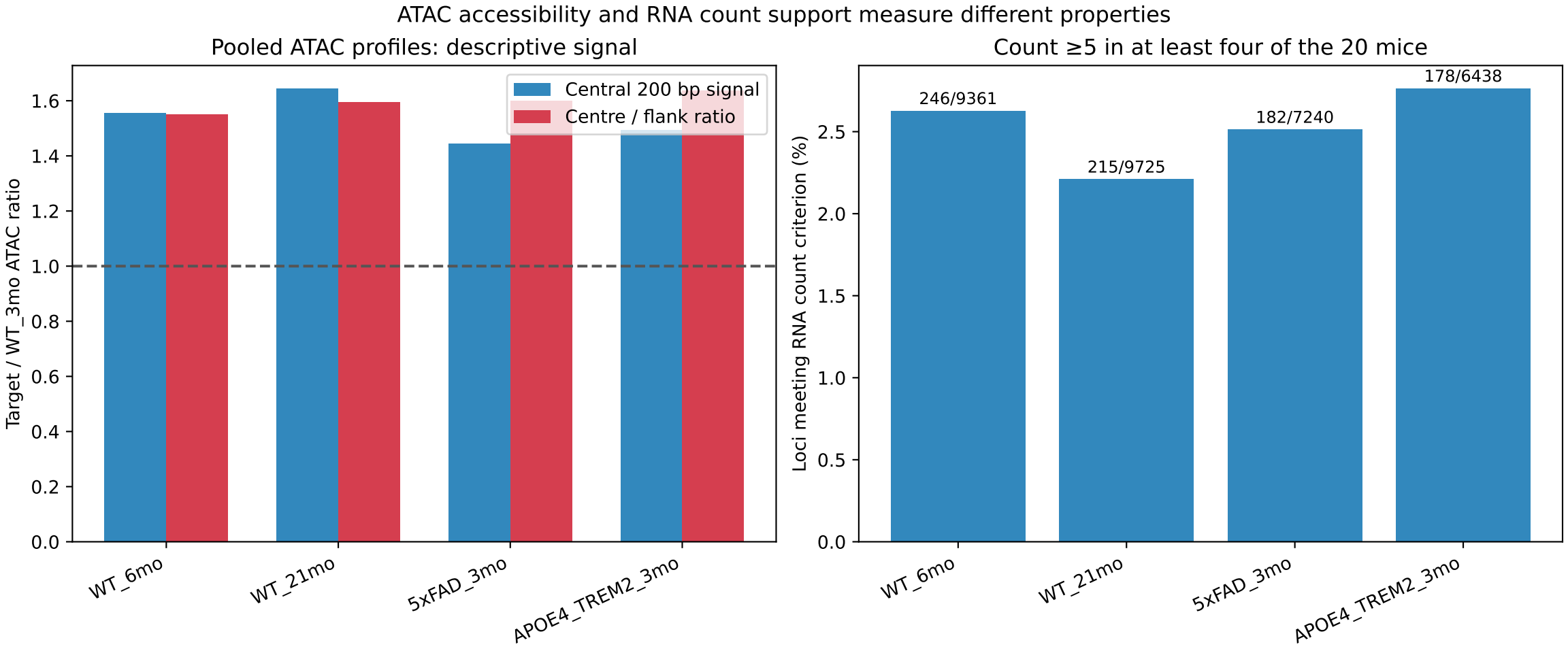

Sex-adjusted negative-binomial models: same fixed target-only TE sets
20 mice, 4 per condition; bars: DESeq2 95% Wald confidence intervals; blue points: edgeR QL estimates
Gene-derived TMM exposure
TE-derived TMM exposure
edgeR
DESeq2
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo
edgeR q=0.993
DESeq2 q=0.997
edgeR q=0.801
DESeq2 q=0.653
edgeR q=0.993
DESeq2 q=0.997
edgeR q=0.993
DESeq2 q=0.997
edgeR q=0.987
DESeq2 q=0.976
edgeR q=0.987
DESeq2 q=0.976
edgeR q=0.687
DESeq2 q=0.596
edgeR q=0.987
DESeq2 q=0.976
0.8
0.9
1.0
1.1
1.2
1.3
1.4
Estimated target / WT_3mo RNA fold change


Model-class and sample-sensitivity audit; gene-derived exposure and sex adjustment
Leave-one-mouse-out prediction
Mean held-out negative log likelihood (lower is better)
Poisson
Negative binomial
0
2
4
6
8
10
12
14
16
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo
Effect sensitivity to omitting one mouse
Estimated RNA log2 fold change
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo


ATAC accessibility and RNA count support measure different properties
Pooled ATAC profiles: descriptive signal
Central 200 bp signal
Centre / flank ratio
Target / WT_3mo ATAC ratio
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo
Count ≥5 in at least four of the 20 mice
Loci meeting RNA count criterion (%)
0.0
0.5
1.0
1.5
2.0
2.5
246/9361
215/9725
182/7240
178/6438
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo